COMSATS University Islamabad
Department of Computer Science

Course Information

Course Code: **CSC354**	Course Title: **Machine Learning**
Credit Hours: **3(3,0)**	Lecture Hours/Week: **3**
Lab Hours/Week: **0**	Pre-Requisites: **None**

Text and Reference Books

Textbooks:

1. Introduction to Machine Learning, Ethem Alpaydin, MIT Press, 2010.
2. Machine Learning, Tom, M., McGraw Hill, 1997.

Reference Books:

1. Hands on Machine Learning with Scikit-Learn and TensorFlow, Aurelien Geron, O`Reilly Media, 2017.
2. Deep Learning with PyTorch – Essential Excerpts, Eli Stevens, Luca Antiga, Thomas Viehmann, Manning Publications, 2009.
3. Pattern Recognition and Machine Learning, Bishop, C., Springer-Verlag, 2007.

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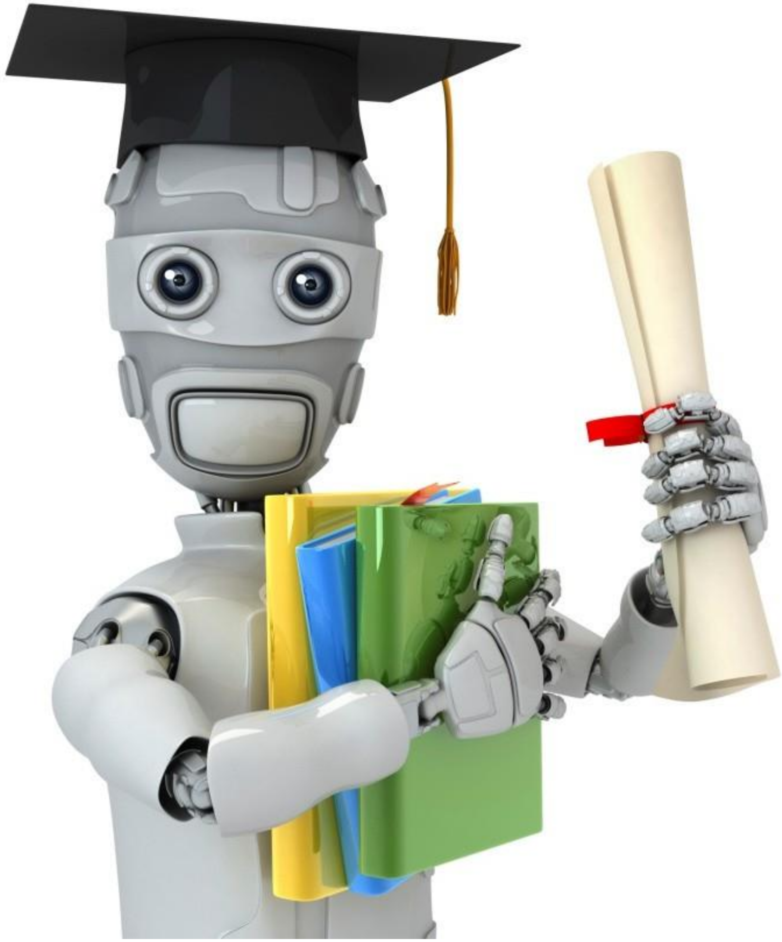
- Zero tolerance for any misconduct and misbehavior
- Zero tolerance for cheating, if student found with any cheating material in quiz/exam will be marked zero
- Copied assignment will be marked zero
- Cross talk with class-fellows will result in strict disciplinary action
- Be in class on time, students who arrive after 5 min will be marked absent
- Submit assignments on portal on time, late assignment wouldn't be considered, and anything send via (WhatsApp etc) wouldn't be considered
- Emails or any message should be sent within working hours rest would not be entertained send email on **(zeenatzulfiqar@cuiatd.edu.pk)**



Machine Learning

SIMULATION TECHNIQUE

Monte Carlo

Simulation

- **What is Simulation?**
- **Simulation** is the process of creating a **model of a real-world system** and running it repeatedly to observe its behavior and predict possible outcomes **without affecting the real system**.

Suppose a supermarket wants to know **how many customers may visit tomorrow**.

- Instead of waiting until tomorrow, it can:
- Use past customer data.
- Create a model.
- Run many virtual experiments.
- This process is called **simulation**

Monte Carlo

- **What is Monte Carlo?**
- A simulation-based algorithm.
- Uses **random sampling**.
- Produces **approximate** solutions.
- Accuracy improves as the number of simulations increases.

Steps of Monte Carlo Simulation

- Define the problem.
- Collect historical data.
- Calculate probabilities.
- Calculate cumulative probabilities.
- Assign random number intervals.
- Generate random numbers.
- Apply the simulation model.
- Record the results.
- Calculate the final estimate.
- Interpret the results.

Monte Carlo Simulation Technique

- A **bookstore** wants to estimate the number of books it will sell tomorrow using the **Monte Carlo Simulation Technique**. The manager collected sales data for the previous **100 days**.
- The following random numbers are generated for the next **10 days**:
- **08, 27, 63, 91, 46, 74, 12, 55, 81, 39**
- **Using the Monte Carlo Simulation Technique, estimate the average number of books sold per day.**

Books Sold per Day	Frequency
20	15
30	35
40	30
50	20
Total	**100**

Step 1: Calculate the Probability

Probability = Frequency ÷ Total Frequency

Books Sold	Frequency	Probability
20	15	15/100 = **0.15**
30	35	35/100 = **0.35**
40	30	30/100 = **0.30**
50	20	20/100 = **0.20**
Total	100	**1.00**

Step 2: Calculate the Cumulative Probability

Previous Cumulative Probability + Current Probability

Books Sold	Frequency	Probability (Frequency ÷ 100)	Cumulative Probability (Running Total)
20	15	15/100 = **0.15**	**0.15**
30	35	35/100 = **0.35**	**0.15 + 0.35 = 0.50**
40	30	30/100 = **0.30**	**0.50 + 0.30 = 0.80**
50	20	20/100 = **0.20**	**0.80 + 0.20 = 1.00**
Total	**100**	**1.00**	**1.00**

Step 3: Assign Random Number Intervals

Books Sold	Probability	Probability × 100	Random Numbers Needed	Random Number Interval	Explanation
20	0.15	0.15 × 100 = **15**	15	**00–14**	Start from **00**. Since 15 numbers are needed, assign **00 to 14** (15 numbers).
30	0.35	0.35 × 100 = **35**	35	**15–49**	Previous interval ended at **14**, so start from **15**. Assign **15 to 49** (35 numbers).
40	0.30	0.30 × 100 = **30**	30	**50–79**	Previous interval ended at **49**, so start from **50**. Assign **50 to 79** (30 numbers).
50	0.20	0.20 × 100 = **20**	20	**80–99**	Previous interval ended at **79**, so start from **80**. Assign **80 to 99** (20 numbers).
Total	**1.00**	**100**	**100**	**00–99**	All **100 random numbers (00–99)** have been assigned.

Step 4: Simulate Using Random Numbers

Day	Random Number	Check the Interval	Books Sold	Explanation
1	**08**	08 lies between **00–14**	**20**	Therefore, books sold = **20**.
2	**27**	27 lies between **15–49**	**30**	Therefore, books sold = **30**.
3	**63**	63 lies between **50–79**	**40**	Therefore, books sold = **40**.
4	**91**	91 lies between **80–99**	**50**	Therefore, books sold = **50**.
5	**46**	46 lies between **15–49**	**30**	Therefore, books sold = **30**.
6	**74**	74 lies between **50–79**	**40**	Therefore, books sold = **40**.
7	**12**	12 lies between **00–14**	**20**	Therefore, books sold = **20**.
8	**55**	55 lies between **50–79**	**40**	Therefore, books sold = **40**.
9	**81**	81 lies between **80–99**	**50**	Therefore, books sold = **50**.
10	**39**	39 lies between **15–49**	**30**	Therefore, books sold = **30**.

Step 5: Calculate the Total Books Sold

- Total books sold
- = 20 + 30 + 40 + 50 + 30 + 40 + 20 + 40 + 50 + 30
- **= 350 books**

Step 6: Calculate the Average

- Average=Number of Simulated Days Total Books Sold
- $= \frac{350}{10}$
- $= 35$

Advantages

- Easy to understand.
- Simple to implement.
- Handles uncertainty.
- Solves complex problems.
- Saves time and cost.
- Supports decision-making.
- Can simulate future events.
- More simulations improve accuracy.

Disadvantages

- Produces approximate results.
- Requires many simulations.
- Can be computationally expensive.
- Depends on the quality of input data.
- Results may vary because of randomness.

Applications

- **Artificial Intelligence**
- Decision-making
- Monte Carlo Tree Search (MCTS)
- Game playing (Chess, Go)
- **Machine Learning**
- Bayesian Learning
- Uncertainty estimation
- Model evaluation

Applications

- **Finance**
- Stock price prediction
- Investment risk analysis
- **Business**
- Demand forecasting
- Inventory management
- Sales prediction
- **Engineering**
- Project planning
- Reliability analysis
- **Healthcare**
- Disease spread modeling
- Hospital resource planning